Activity: CIA in Action

#	Scenario	Which Part(s) of CIA?	Explain Your Reasoning
1	A hospital is hit by ransomware, locking doctors out of patient records before surgery.	Availability	Data not accessible
2	An e-commerce site is hacked, and customer credit card numbers are posted online.	Confidentiality	Customers data is no longer private
3	A cloud backup fails, and the company can't recover files after a server crash.	Availability	Files not available after crash
4	A student accidentally changes someone else's grades in the system due to a bug.	Integrity	Grade was altered
5	A company uses strong encryption to protect emails sent to clients.	Confidentiality/Integrity	Encryption keeps emails private, and unaltered
6	A government website goes offline for hours due to a denial-of-service (DoS) attack.	Availability	Website not available
7	An HR manager leaves their unlocked laptop in a public place, exposing employee payroll data.	Confidentiality	Employee payroll data is is potentially not private anymore
8	A finance app verifies all transactions using checksums to prevent tampering.	Integrity	App keeps transactions accurate
9	After a natural disaster, only the main office can access the payroll system.	Availability	Payroll system not available to everyone
10	A healthcare app limits access to patient data based on staff roles.	Confidentiality/Integrity	App limits users that can view data, which also maintains accuracy

1. In this exercise, I saw Availability, and Confidentiality the most (4 times each). I believe that this is because there are so many reasons that Availability can go away, from natural disasters to DoS attacks. As far as Confidentiality, I think when a website, or entity is compromised, all the data within it is now made not private. This is the case with the Solar Winds, and Target attack case studies.

2. There were two scenarios that involved Confidentiality and Integrity. They connected because (in the case of 5 the encryption question) the encryption serves to keep the email private, and in doing so also maintains accuracy. The healthcare app used role based access which limits who has access to the patient data, which means less people potentially altering, or viewing the data.

3. These principles not only apply to your school, job, or personal data, but to most people's lives. Everyone has a SS#, or some form of ID, and that data is stored in a database somewhere. The core principles of Confidentiality, Integrity, and Availability allow most people to function in today's society. It's how you pay for things, how you are identified, and how you're able to do your job. Without those principles you become susceptible to ID theft, fraud, and many other nefarious problems.